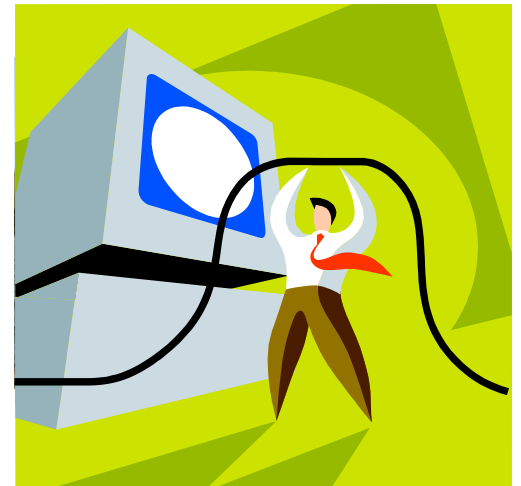


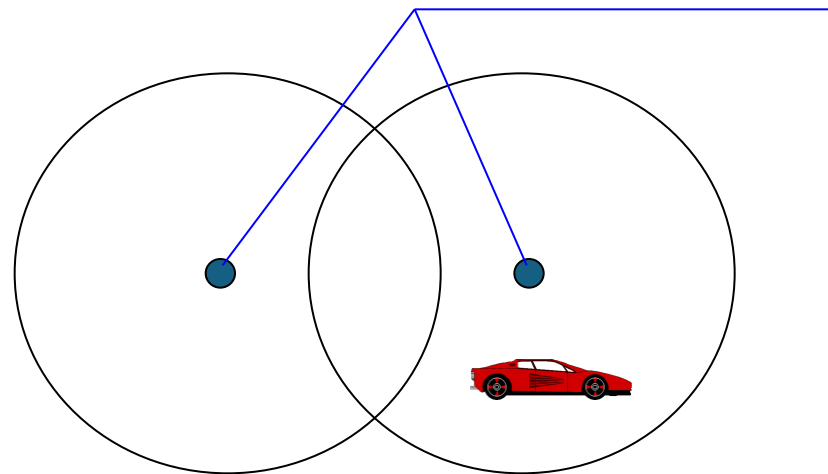
Mobile Networks

Connections and structures



Public cellular network

- Access network with radio link
 - Space is divided in cells with a base station
 - Mobile Node (MN) can work when changing between cells



Cell coverage size is

- Highly variable
- Depends on the technology
- Depends on the number of users

Cells

Advantages:

- > capacity
- > # users
- < power
- > robustness (distributed system)

Each cell locally takes care of interference, coverage area, etc...

• Disadvantages

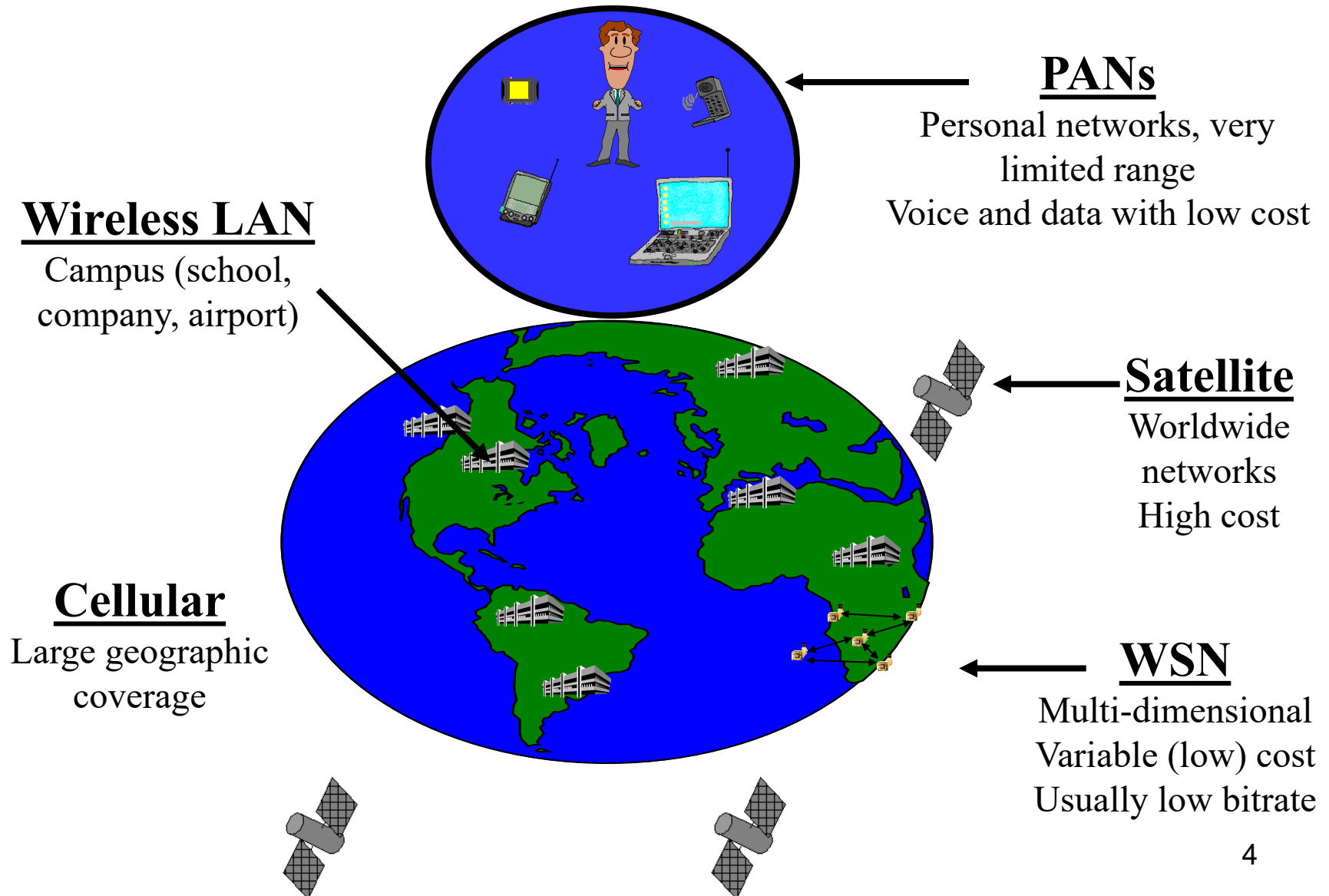
- Uses cabled network between cells
- Many handovers
- Interference between cells

• Fundamental:

Cell dimensioning

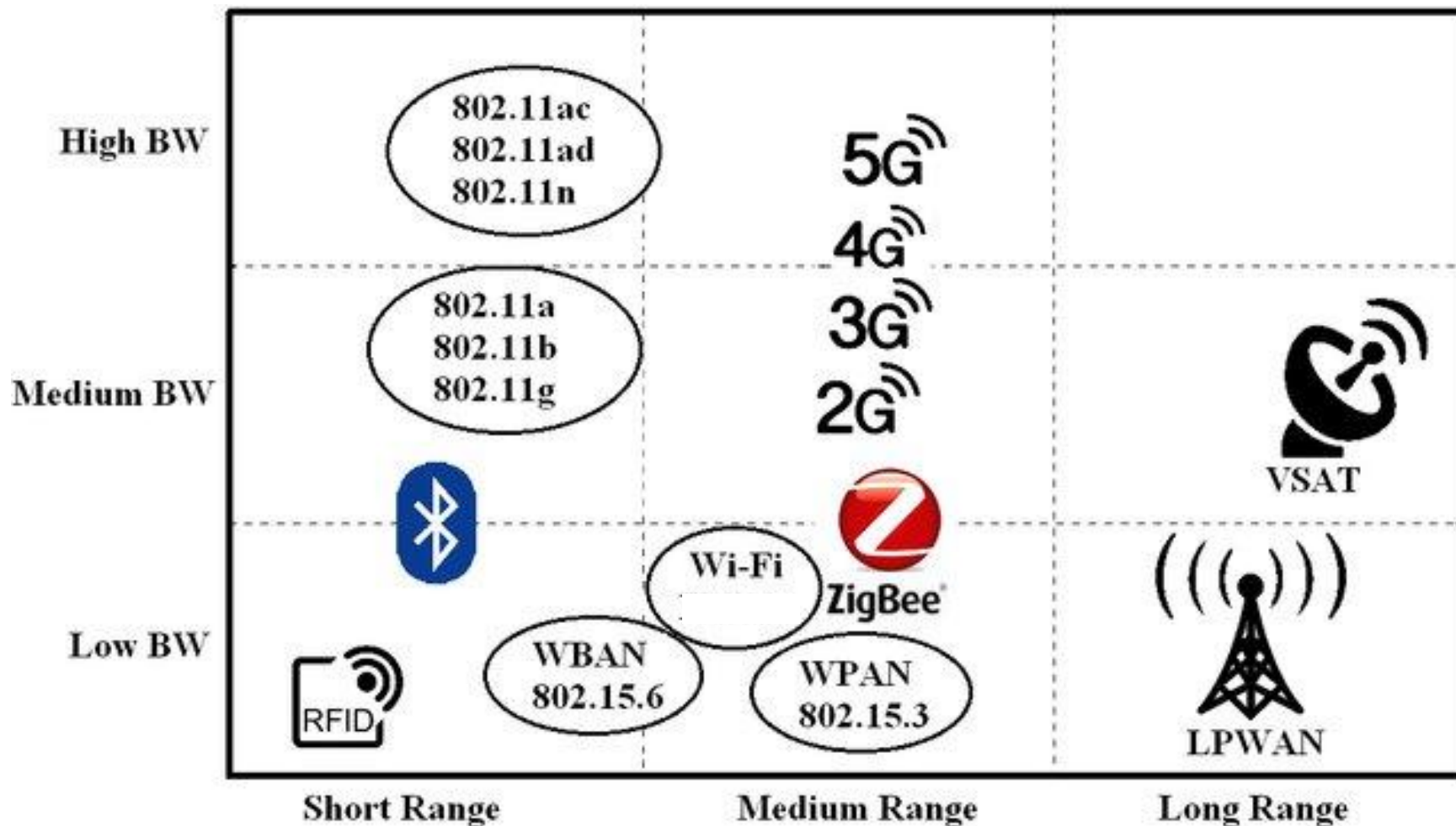
- Length of the cell
- Frequency re-utilization
- Channel reservation

Types of Wireless networks



Comparison Between Wireless Technologies

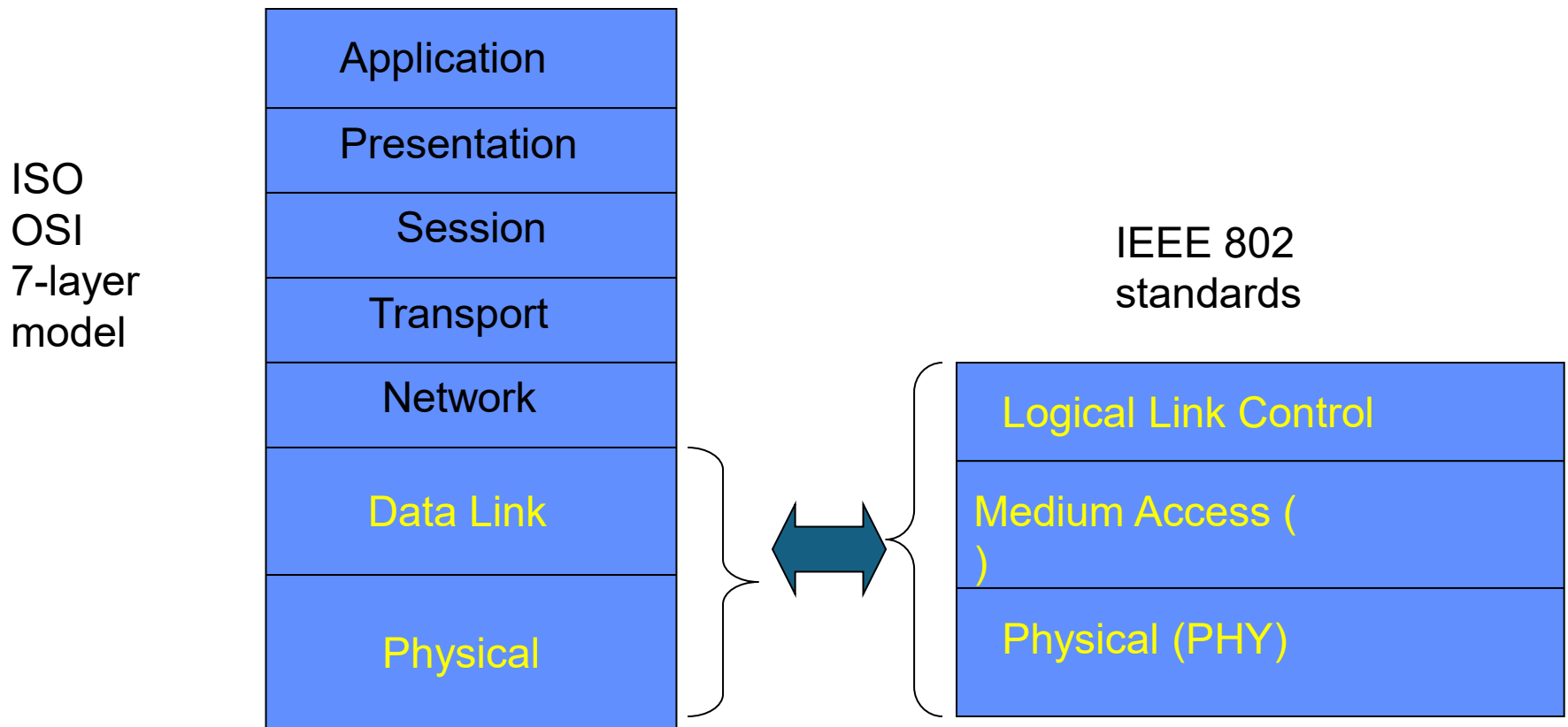
Tradeoff between data rate and range



Standardization of Wireless Networks

- Wireless networks are standardized by IEEE.
- Under 802 LAN MAN standards committee.

LAN – Local Area Network
MAN – Metro Area Network



802.11

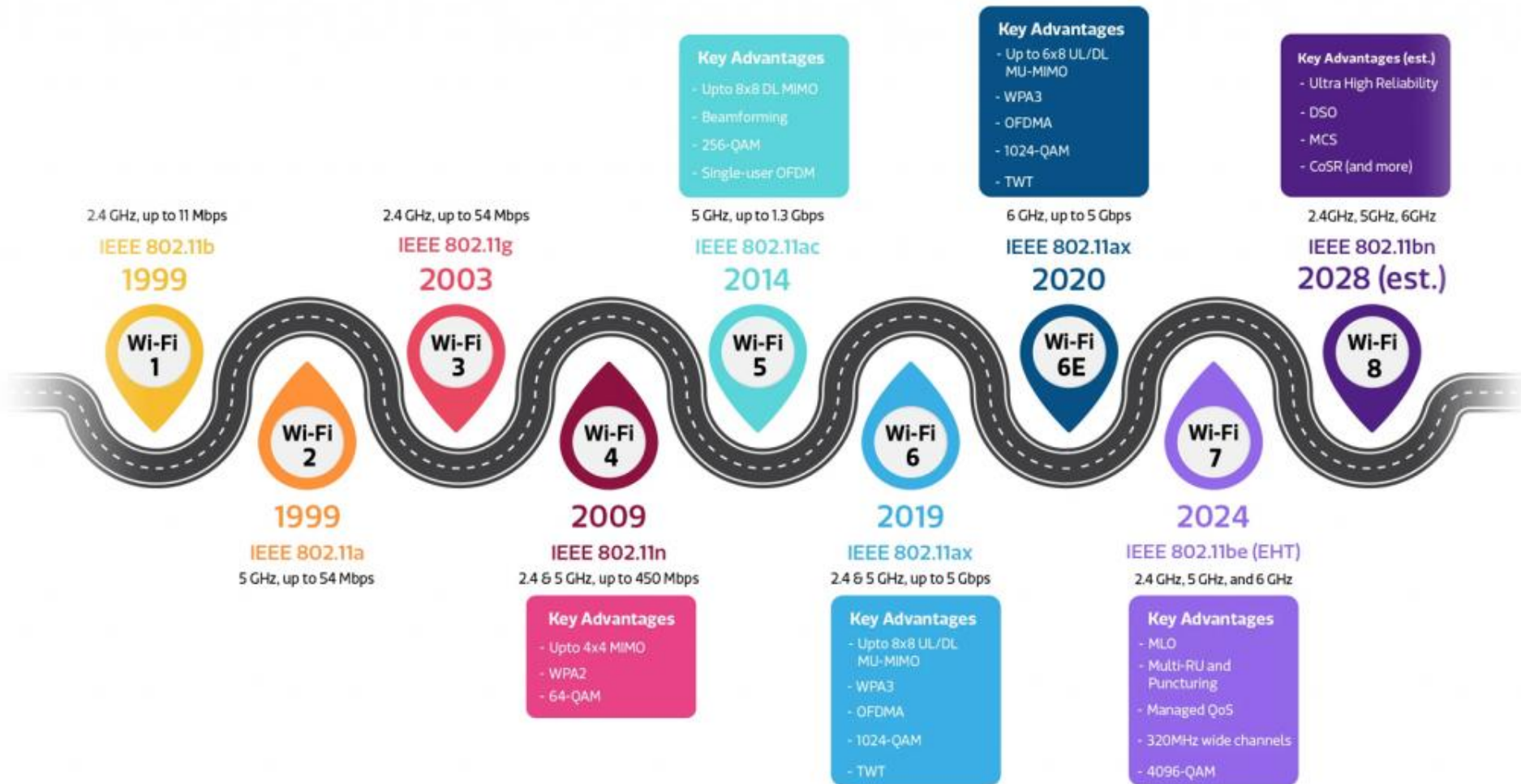
Outline

- 802.11 standard
 - Physical layer
- MAC
 - DCF – Distributed Coordination Function
 - PCF – Point Coordination Function
- Advanced MAC functions

Historic IEEE 802.11 standard

- Local Wireless Network (WLAN)
- Includes Medium Access Control (MAC)
- Includes(d) five physical layers (PHY)
 - Frequency Hopping Spread Spectrum
 - Direct Sequence Spread Spectrum
 - infrared
 - 11 Mbps - 2.4 GHz
 - 54 Mbps - 5 GHz
 - Early efforts divided in three standards:
 - 802.11
 - 802.11a
 - 802.11b

Wi-Fi Evolution



Historic IEEE 802.11 Family

Protocol	Release Data	Freq.	Rate (typical)	Rate (max)	Range (indoor)
Legacy	1997	2.4 GHz	1 Mbps	2Mbps	?
802.11a	1999	5 GHz	25 Mbps	54 Mbps	~30 m
802.11b	1999	2.4 GHz	6.5 Mbps	11 Mbps	~30 m
802.11g	2003	2.4 GHz	25 Mbps	54 Mbps	~30 m
802.11n	2008	2.4/5 GHz	200 Mbps	600 Mbps	~50 m
802.11ac	2014	5 GHz	600Mbps	3.5 Gbps	~35m
802.11ax (Wi-Fi 6)	2021	2.4/5 GHz	130 (2.4 GHz) 400-800Mbps (5GHz)	10 Gbps	~30m
802.11be (Wi-Fi 7)	2024	2.4/5/6 GHz	2-4Gbps (laptop) , 5-18 PC's	40 Gbps	Similar to Wi-Fi 6
802.11ay	2021	60 GHz	20 Gbps	20-40 Gbps	300-500m
802.11bn	2028?	2.4/5/6 GHz	similar	Similar	similar

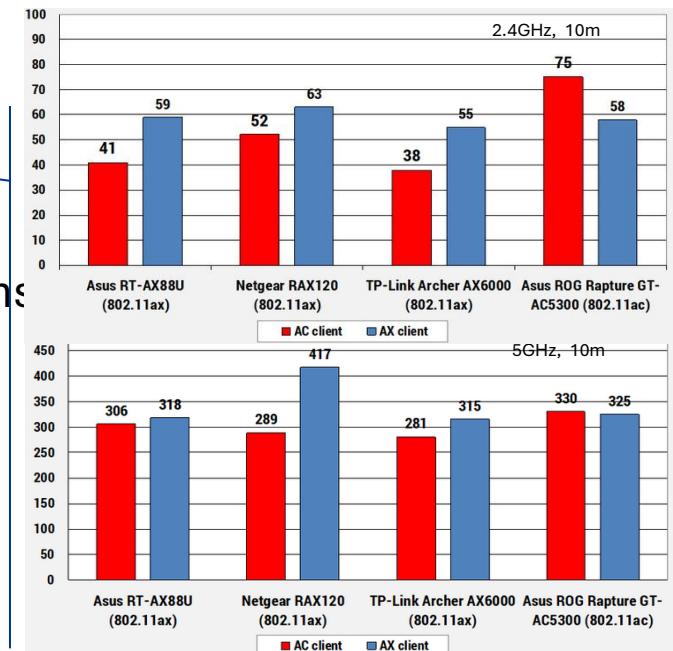
New 802.11 Radio technologies

Current recent innovations being deployed:

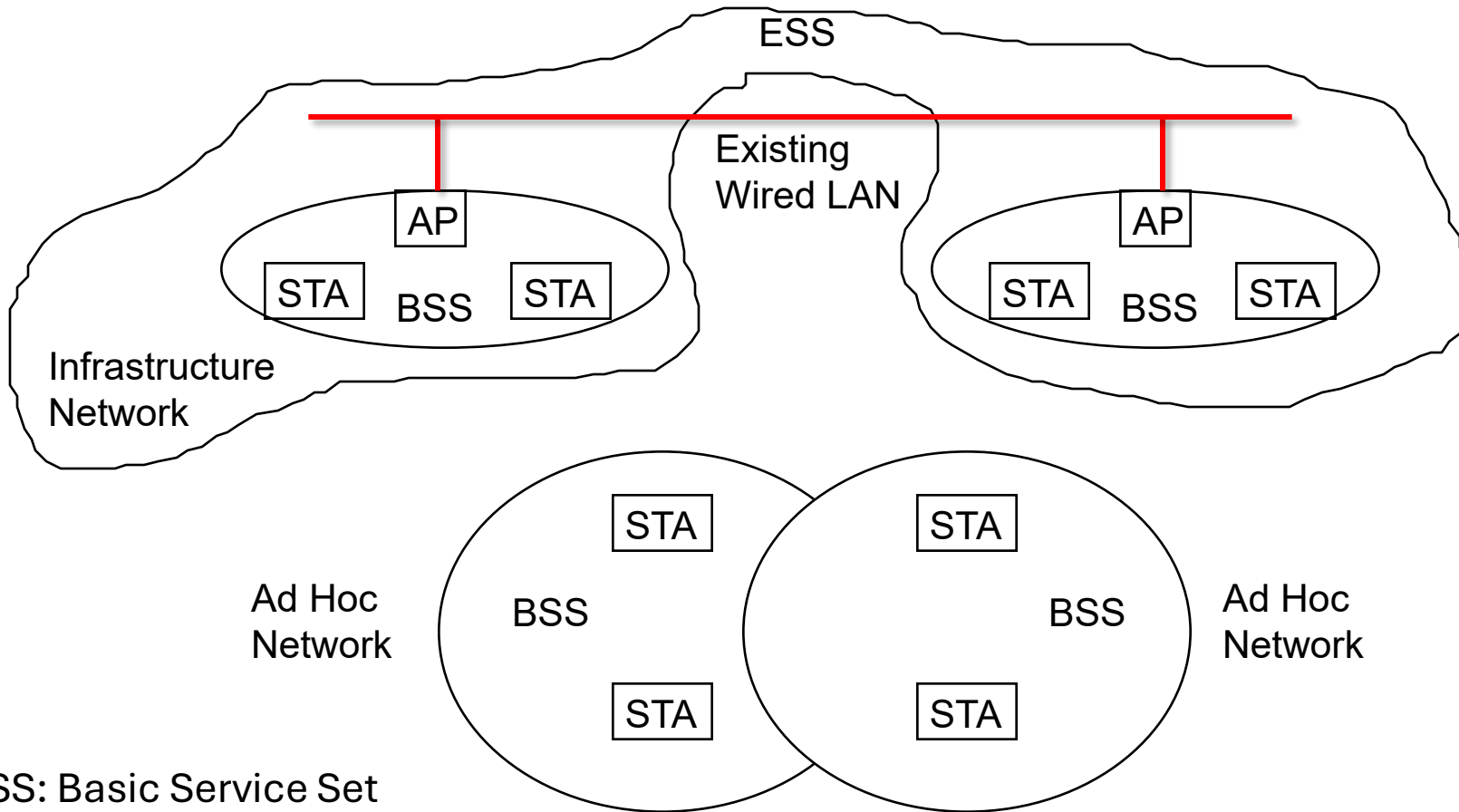
- 802.11ax – Increased throughput in 2.4, 5 (and 6) GHz bands. Increased efficiency.

WiFi6

- 802.11ay – Support for 20 Gbps in 60 GHz band.
- 802.11az – 2nd generation positioning features.
- 802.11ba – Wake up radio. Low power IoT applications
- 802.11bb – Light Communications
- 802.11bc – Enhanced Broadcast Service
- 802.11bd – Enhancements for Next Generation V2X
- 802.11be – Extremely High Throughput
- 802.11bf – WLAN Sensing [pending approval]



802.11 Architecture



BSS: Basic Service Set

ESS: Extended Service Set

DS: Distribution System

Components

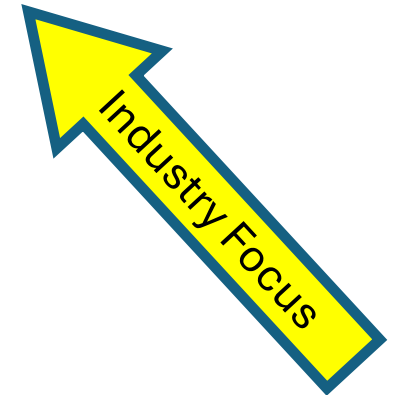
- Station (STA) — Mobile Terminal
- Access Point (AP) — STA are connected to Access Points (infrastructured networks)
- Basic Service Set (BSS) — STA and AP with the same coverage and connectivity area create a BSS.
- Extended Service Set (ESS) — Multiple BSSs connected via the APs create an ESS.
- Distribution System (DS) - Contains the entity that interconnects APs

Distribution System (DS)

- The Distribution system interconnects multiple BSSs
- 802.11 standard **logically separates** the wireless medium from the distribution system – it does not preclude, nor demand, that the multiple media be same or different
- An Access Point (AP) is a STA that provides access to the DS by providing DS services in addition to acting as a STA.
- Data moves between BSS and the DS via an AP
- The DS and BSSs allow 802.11 to create a wireless network of arbitrary size and complexity called the **Extended Service Set** network (ESS)

Infrastructure vs Ad Hoc Mode

- Infrastructure mode: stations communicate with one or more access points which are connected to the wired infrastructure
 - What is deployed in practice
- Two modes of operation:
 - Distributed Control Functions - DCF
 - Point Control Functions – PCF
 - PCF is rarely used - inefficient
- Alternative is “ad hoc” mode: multi-hop, assumes no infrastructure
 - Rarely used, e.g. military
 - Hot research topic!



What about Ad Hoc?

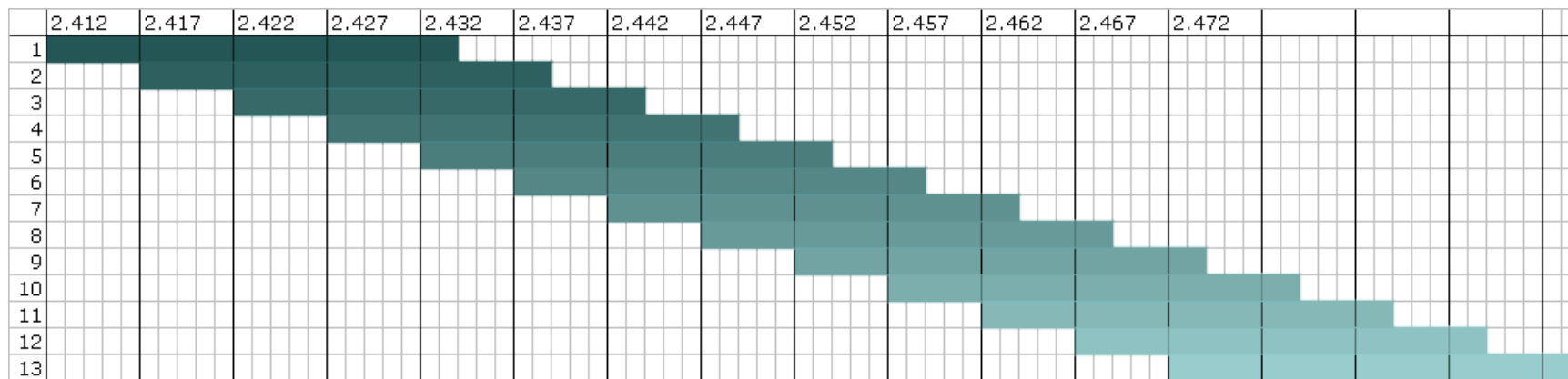
- Ad-hoc mode: no fixed network infrastructure
 - Based on an Independent BSS
 - A wireless endpoint sends and all nodes within range can pick up signal
 - Each packet carries destination and source address
 - Effectively need to implement a “network layer”
 - How do know who is in the network?
 - Routing?
 - Security?

Outline

- 802.11 standard
 - Physical layer
- MAC
 - DCF
 - PCF
- Advanced MAC functions

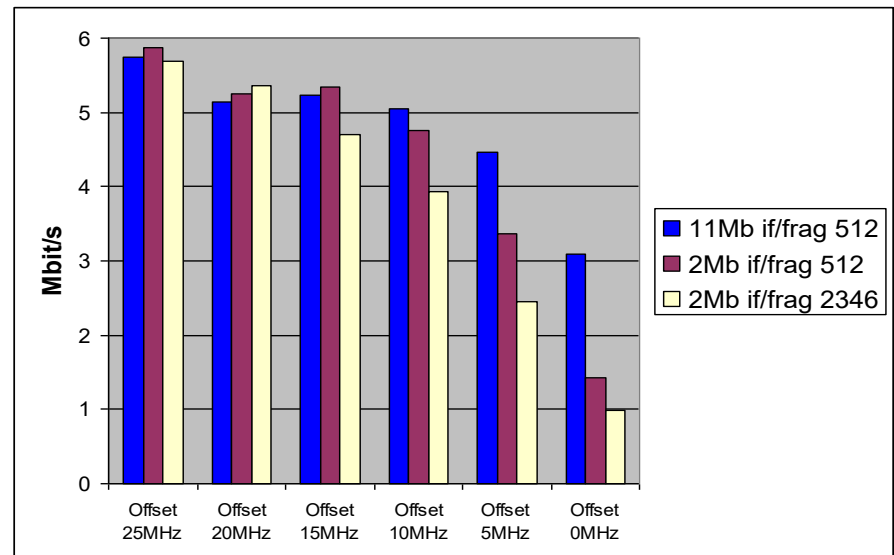
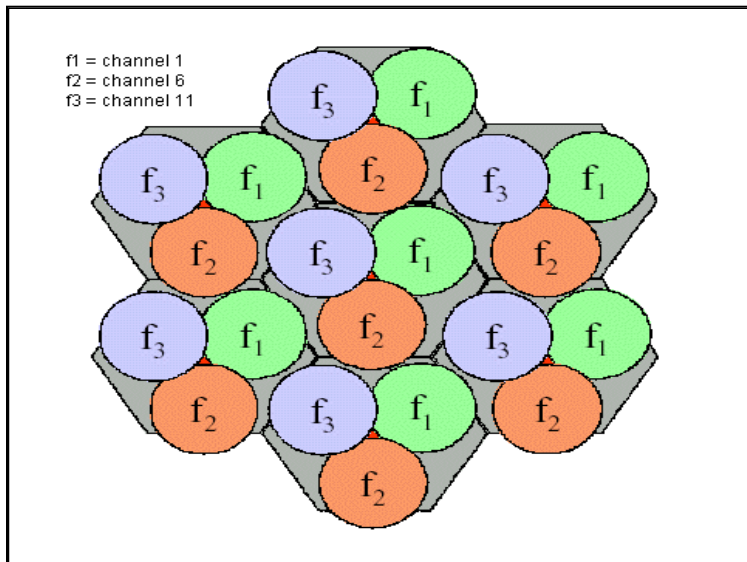
802.11 Channels (2.4GHz)

- The frequency is divided in channels
- In the UK and most of EU: 13 channels, 5MHz apart, 2.412 – 2.472 GHz
- In the US: only 11 channels
- Each channel is 22 MHz
- Significant overlap
- Best channels are 1, 6 and 11



Frequency planning

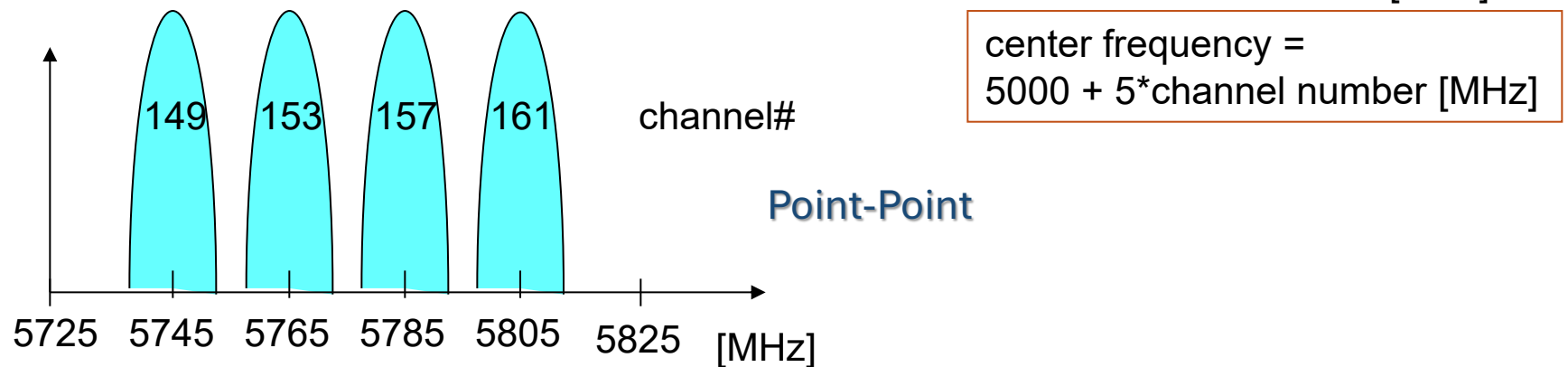
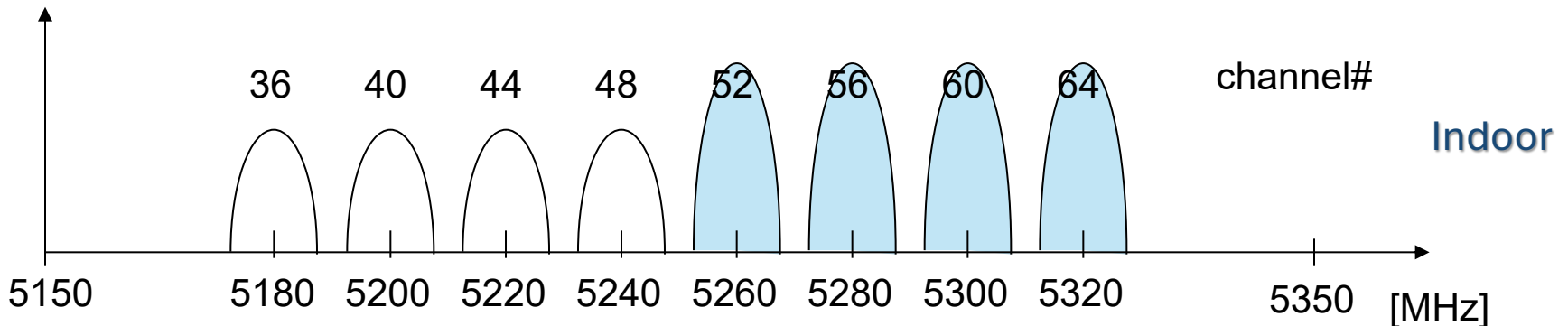
- Interference from other WLAN systems or cells
- IEEE 802.11 operates at uncontrolled ISM band
- 14 channels of 802.11 are overlapping, only 3 channels are disjointed. For example Ch1, 6, 11
- Throughput decreases with less channel spacing
- A example of frequency allocation in multi-cell network



802.11 (5GHz)

- Uses frequency division in the 5.2 and 5.7 GHz bands
- What are the benefits?
 - Greater bandwidth
 - Less potential interference (5GHz)
 - More non-overlapping channels
- But does not provide interoperability
 - Interoperability at chipset level

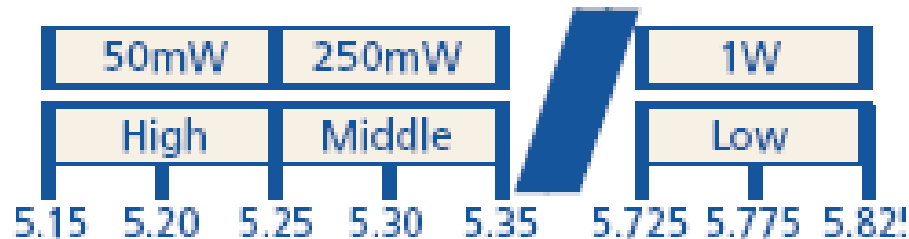
Example: 802.11a Physical Channels



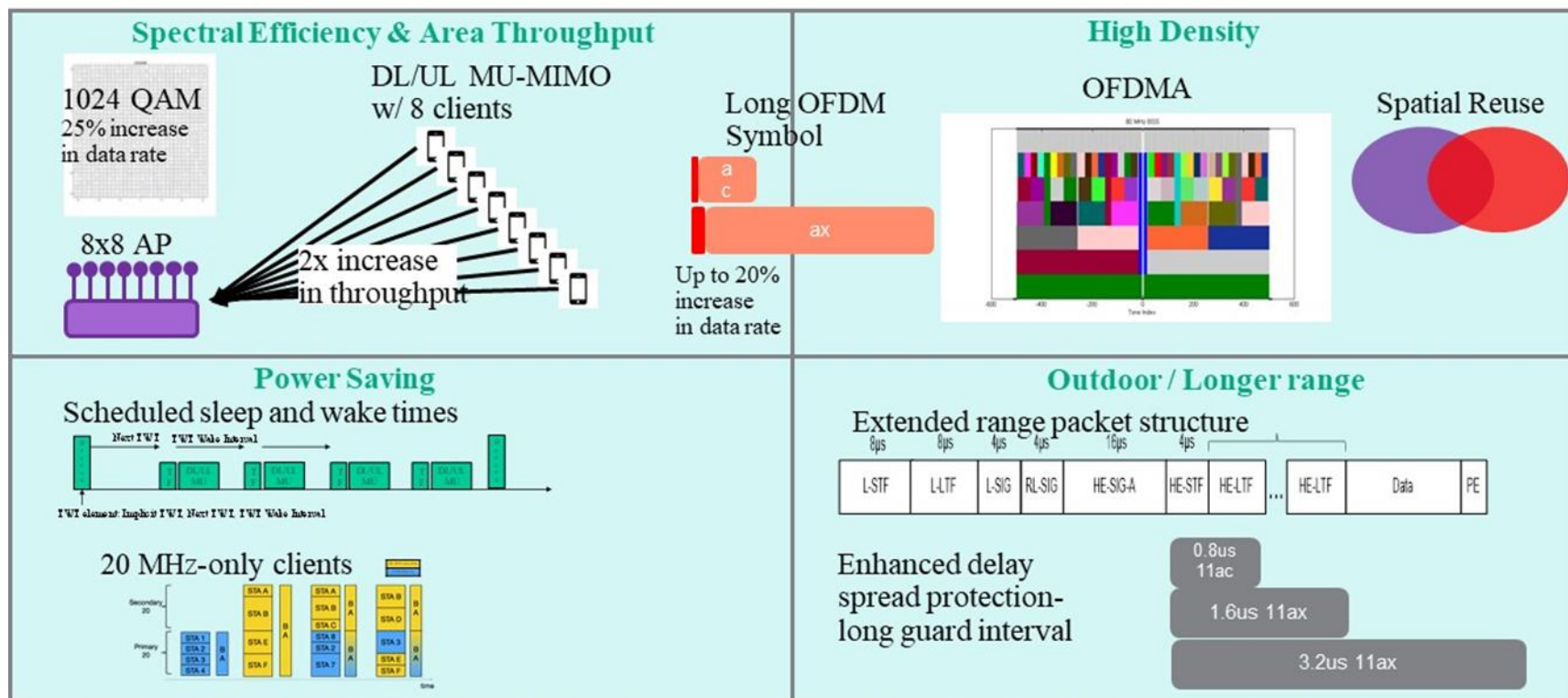
Maximum Power Output

U-NII Band

Frequency (GHz)



WiFi 6 radio layer enhancements



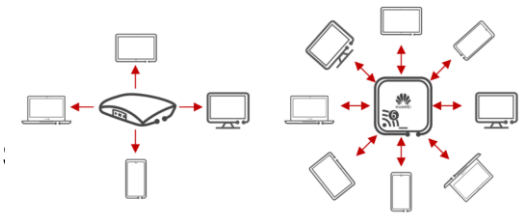
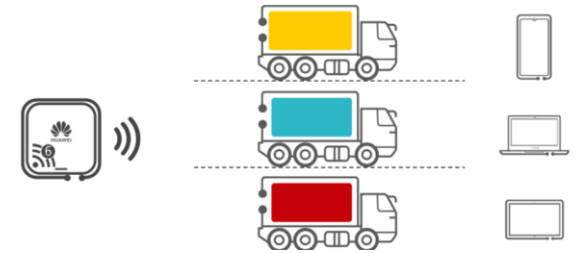
OFDMA – Orthogonal Frequency-division Multiple Access

- Multi-user version of OFDM (Orthogonal frequency-division Multiplexing)
- Divides channel resources into multiple Resource Units (RUs)
- Different users are allocated these RUs
- Data of multiple users can be sent on one channel simultaneously
- New in Wi-Fi 6
- So:
 - The AP communicates with multiple users during one transmission period

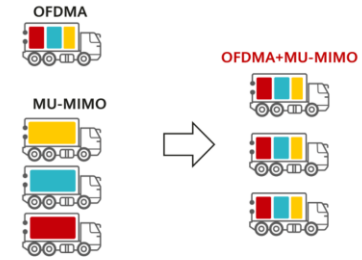


MU-MIMO – Multi-user Multiple-Input and Multiple-Output

- Introduced in Wi-Fi 5
- # antennas in APs is greater than in terminals
 - Unable to make the most out of channel resources
 - E.g.: in 802.11ac, which is only in 5GHz, each spatial stream (1x1 MIMO) has a max PHY rate of 433 Mbps when used with 80 MHz-wide channels
 - Single user transmission
- With MU-MIMO
 - AP communicates with multiple terminals simultaneously
 - Wi-Fi 5: 4x4 DL MU-MIMO (4 * 433 in downlink only)
 - Wi-Fi 6: 8x8 UL/DL MU-MIMO (8 * 433 in uplink/downlink)
- MU-MIMO



OFDMA + MU-MIMO



- MU-MIMO
 - Physically divides network resources to increase capacity and efficiency in high-bandwidth applications (i.e., video streaming and download)
 - Increases spatial stream utilization and effective bandwidth while also lowering latency
 - Prone to impact from terminals
- OFDMA
 - Supports multi-channel transmission in the frequency domain
 - Ideal for low-bandwidth, small-packet applications (e.g., web browsing, IM)
 - Increases spatial stream utilization and queueing time.
 - Stable and resilient to impact from terminals
- MU-MIMO + OFDMA = Complementary operation
 - Optimal resource allocation based on services, via joint scheduling

Wi-Fi 7

- 6 GHz band!
 - In reality, Wi-Fi 6E also had...
 - Maximum channel bandwidth: 320MHz
 - Wi-fi 6: 160MHz
 - Analogy: highways with more lanes
- Quadrature Amplitude Modulation (QAM)
 - Data is represented by combinations of amplitudes, phases or frequencies
 - The encoding scheme determines the number of bits that can be carried in a symbol
 - Wi-Fi 6 uses 1024-QAM (10 bits) ... Wi-Fi 7 used 4096-AQM (12 bits → 1.2x +)
- Multi-link Operation (MLO): 2.4GHz + 5GHz+ 6 GHz
- Peak transmission rate:
 - Wi-fi 6: 9.6Gbps
 - Wi-Fi 7: 23.06Gbps (x2.4 times!)

Wi-Fi 8

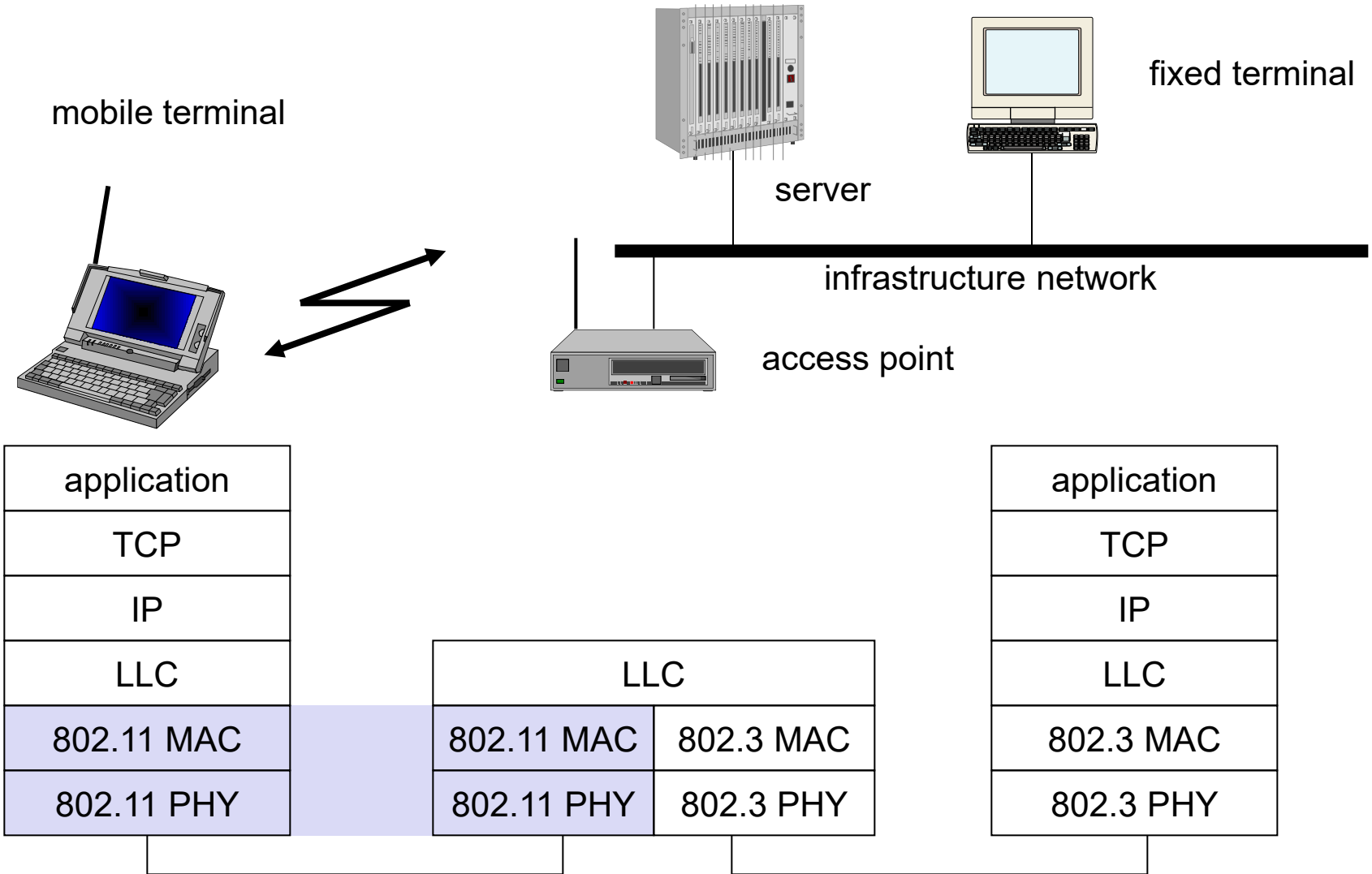
- Multi-AP Coordination
- Channel BW 320 MHz (and more)
- 8192 QAM Modulation (higher quantity of signals modulated)
- mmWave spectrum
- Distributed Multi-Link Operation



Outline

- 802.11 standard
- Physical layer
- MAC
 - DCF
 - PCF
- Advanced MAC functions

802.11- in the TCP/IP stack



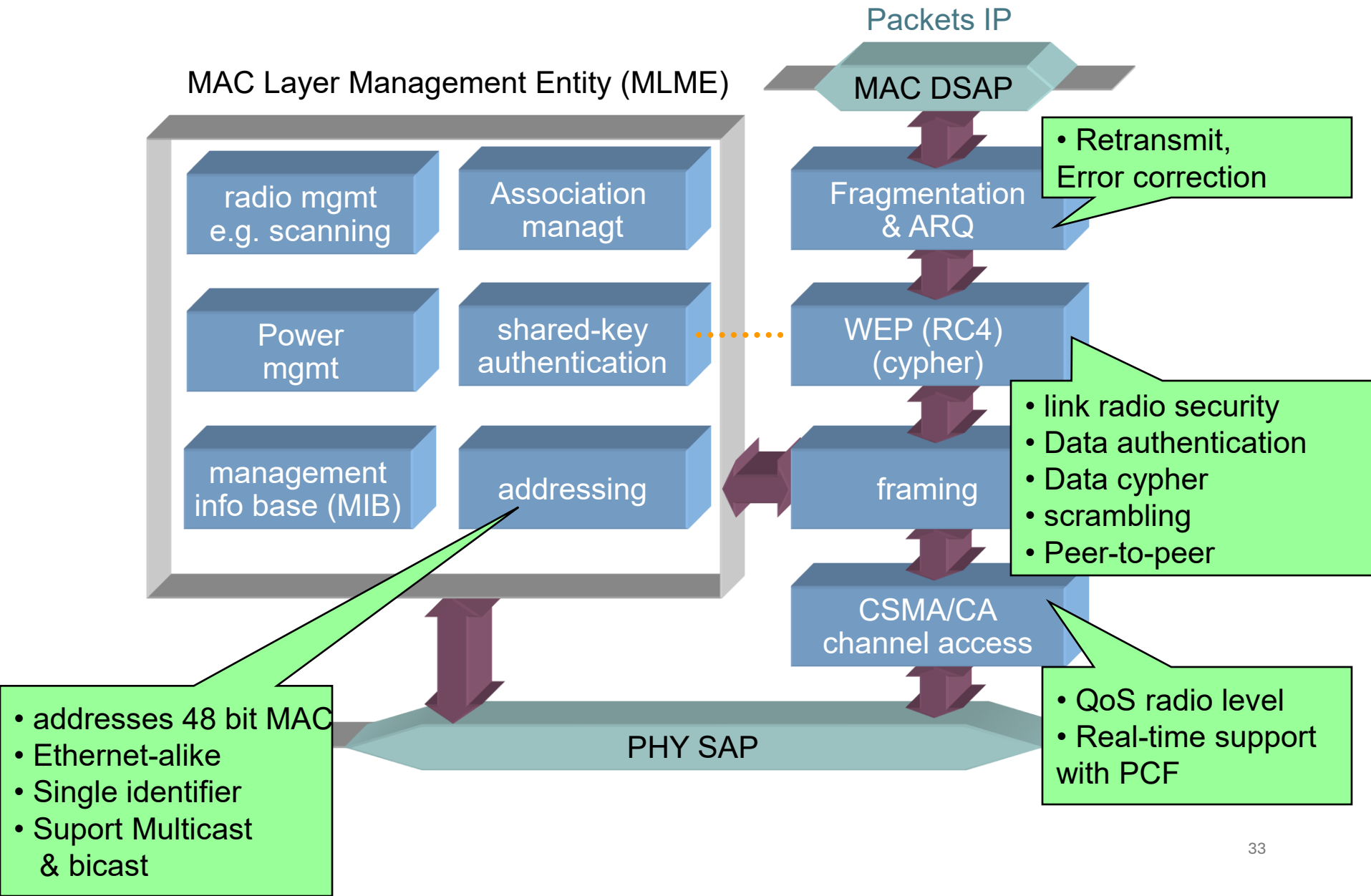
802.11 MAC Overview

- Uses variant of Carrier Sense Multiple Access with Collision Avoidance (CS/MACA)
 - RTS/CTS used for addressing hidden-nodes
- Automatic Repeat Request (ARQ)
 - Error control method for reliability
 - All frames have to be properly ACK, or timeout occurs
- Two operating modes:
 - Infra-structured network (Access point)
 - Ad-Hoc networks (without access point)
- Power saving support
- Wired Equivalent Privacy (WEP)
- MAC management
- Independent of the physical layer or of operating mode

Features of 802.11 MAC protocol

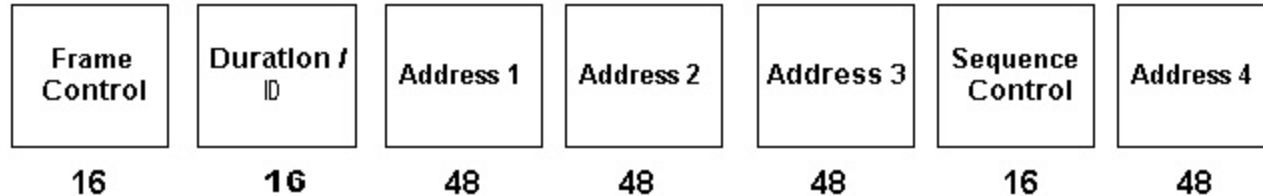
- Fair control access
 - Supports Media Access Control functionalities
 - Addressing
 - CSMA/CA
- Protection of data
 - Error detection (FCS – Frame Check Sequence)
 - Compares number with received values
 - Error correction (ACK frame)
- Reliable data delivery
 - Fragmentation
 - Flow control: stop-and-wait (the next frame is only sent after an ACK from the previous one is received)

MAC IEEE802.11

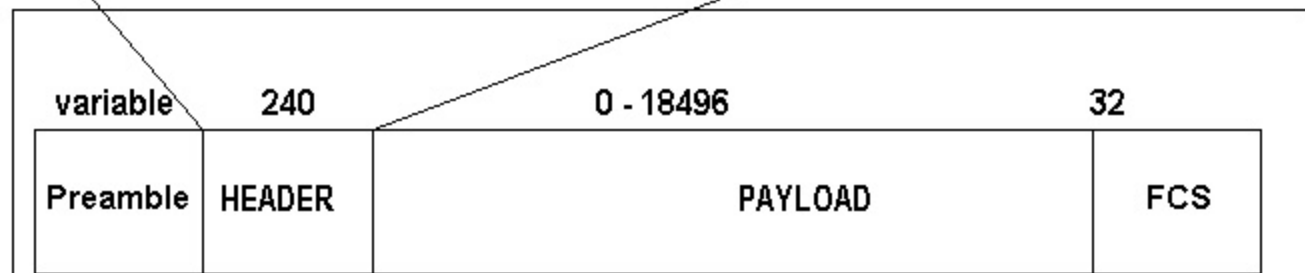


802.11 Frames

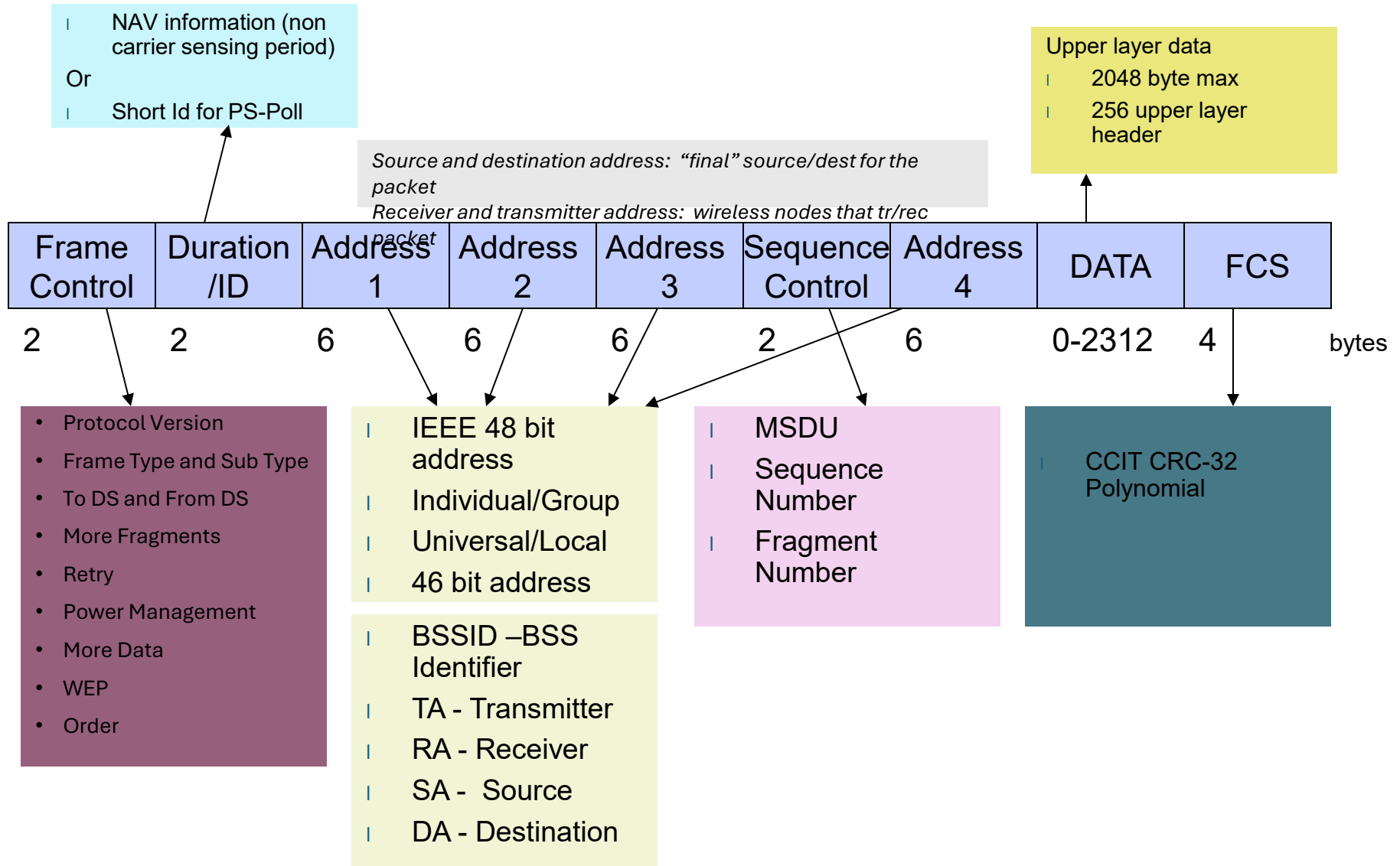
- Three types of frames
 - control: RTS, CTS, ACK
 - Management
 - Data
- Header depends on the frame type



The 240 bit header may be truncated, based on specific frame type



Frame Format



Packet Types

- Type/sub-type field is used to indicate the type of the frame
- Management:
 - Association/Authentication/Beacon
- Control
 - RTS, CTS, CF-end, ACK
- Data
 - Data only, or Data + CF-ACK, or Data + CF-Poll or Data + CF-Poll + CF-ACK

CF → Contention Free

Some More Fields

- Duration/ID: Duration in DCF mode/ID is used in PCF mode
- More Frag: 802.11 supports fragmentation of data
- More Data: In polling mode, station indicates it has more data to send when replying to CF-POLL
- RETRY is 1 if frame is a retransmission;
- WEP (Wired Equivalent Privacy) is 1 if frame is WEP coded
- Power Mgmt is 1 if in Power Save Mode;
- Order = 1 for strictly ordered service

Multi-bit Rate

- 802.11 allows for multiple bit rates
 - Allows for adaptation to channel conditions
 - Specific rates dependent on the version
- Algorithm for selecting the rate is not defined by the standard – left to vendors
- Packets have multi-rate format
 - Different parts of the packet are sent at different rates
- Short vs Long preamble
 - Preamble allows the receiver to synchronize with the transmitter
 - Additional data is added to the header to help check for transmission errors
 - Long
 - Older, requires more data to help check for transmission errors (does it better)
 - Short
 - Less data = faster

Addressing Fields

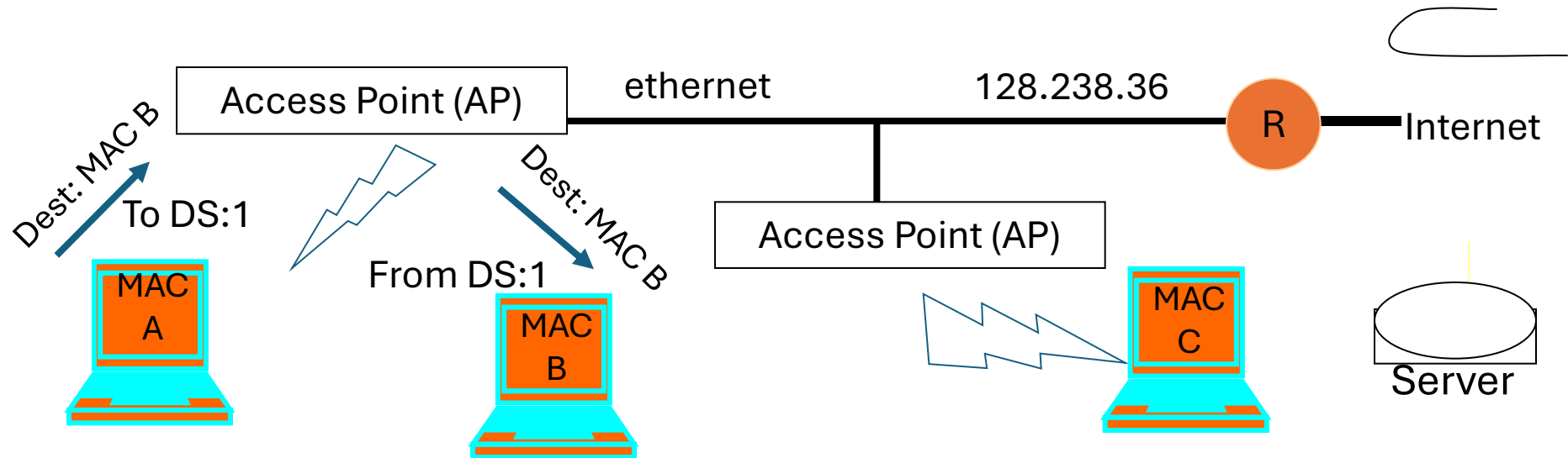
To DS	From DS	Message	Address 1	Address 2	Address 3	Address 4
0	0	station-to-station frames in an IBSS; all mgmt/control frames	DA	SA	BSSID	N/A
0	1	From AP to station	DA	BSSID	SA	N/A
1	0	From station to AP	BSSID	SA	DA	N/A
1	1	From one AP to another in same DS	RA	TA	DA	SA

RA: Receiver Address TA: Transmitter Address
DA: Destination Address SA: Source Address
BSSID: MAC address of AP in an infrastructure BSS

Data Flow Examples

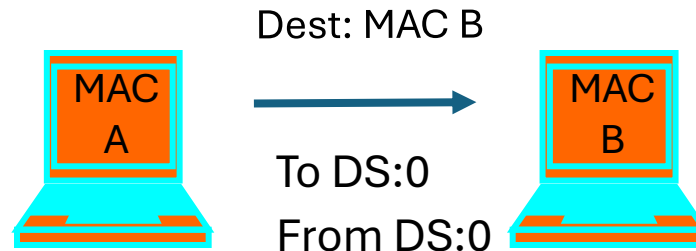
- Case 1: Packet from a station under one AP to another in same AP's coverage area
- Case 2: Packet between stations in an IBSS
- Case 3: Packet from an 802.11 station to a wired server on the Internet
- Case 4: Packet from an Internet server to an 802.11 station

Case 1: Communication Inside BSS



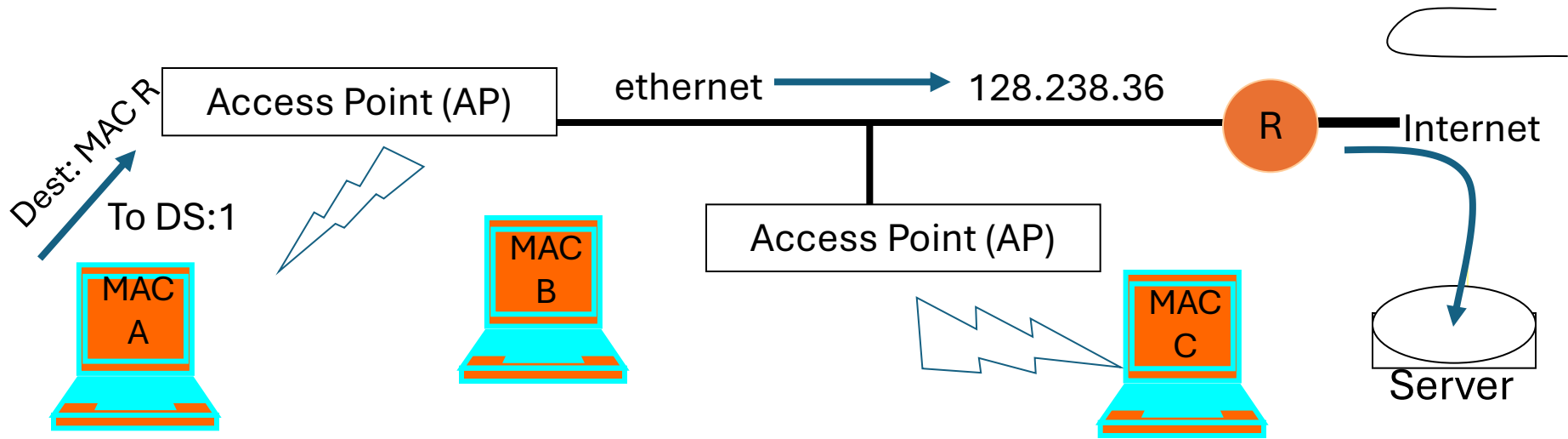
- AP knows which stations are registered with it so it knows when it can send frame directly to the destination

Case 2: Ad Hoc



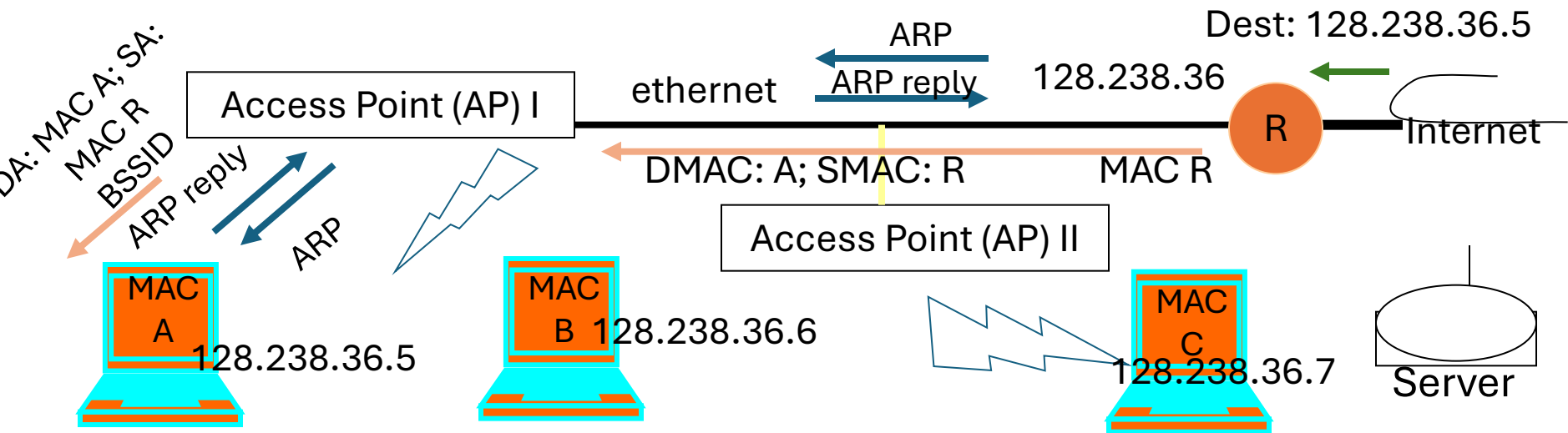
- Direct transmit only in IBSS (Independent BSS), i.e., without AP
- Note:
 - in infrastructure mode (i.e., when AP is present), even if B can hear A, A sends the frame to the AP, and AP relays it to B

Case 3: To the Internet



- MAC A determines IP address of the server (using DNS)
- From the IP address, it determines that server is in a different subnet
- Hence it sets MAC R as DA;
 - Address 1: BSSID, Address 2: MAC A; Address 3: DA
- AP will look at the DA address and send it on the ethernet
 - AP is an 802.11 to ethernet bridge
- Router R will relay it to server

Case 4: From Internet to Station



- Packet arrives at router R – uses ARP to resolve destination IP address
 - AP knows nothing about IP addresses, so it will simply broadcast ARP on its wireless link
 - DA = all ones – broadcast address on the ARP
- MAC A host replies with its MAC address (ARP reply)
 - AP passes on reply to router
- Router sends data packet, which the AP simply forwards because it knows that MAC A is registered

Let's stop here!
Next: Practical Work!